Exploratory Data Analysis of the Titanic Dataset

1. Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to understand passenger characteristics, identify patterns, examine relationships between variables, and analyze survival outcomes.

2. Dataset Description

The Titanic dataset contains passenger information, including age, gender, passenger class, ticket fare, family relationships, and survival status.

3. Statistical Analysis

Statistical analysis was performed using Pandas functions such as `describe()`, `info()`, and `value_counts()`. Missing values were also identified using `isnull().sum()`.

4. Visualizations and Observations

Histogram: Shows the distribution of passenger ages and helps identify common age groups.

Boxplot: Displays the distribution of ticket fares and highlights potential outliers.

Scatterplot: Examines the relationship between passenger age and ticket fare.

Heatmap: Displays correlations between numerical variables.

Pairplot: Compares relationships among age, fare, passenger class, and survival status.

5. Key Findings

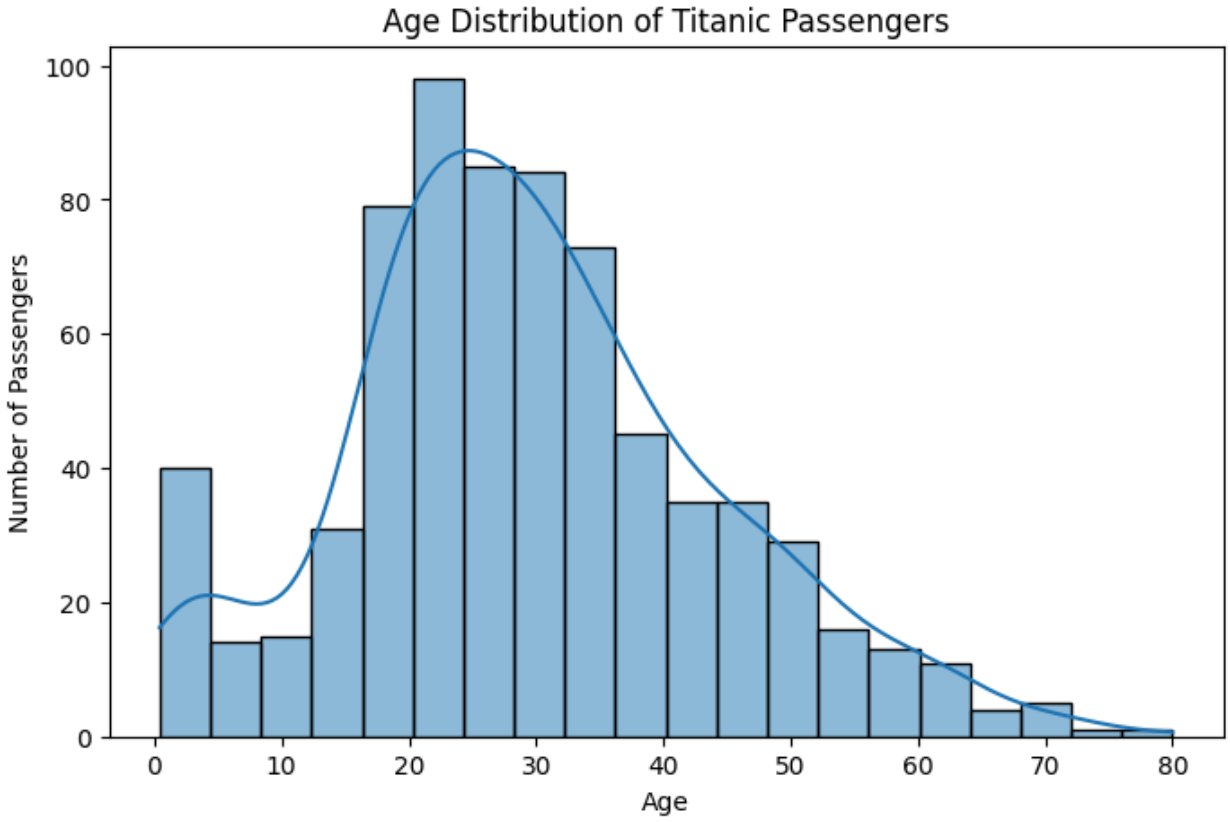
- The dataset contains information about Titanic passengers.
- Passenger ages contain missing values.
- Ticket fares vary across passengers.
- Passenger class, age, gender, and survival status can be examined to identify patterns.
- Correlation analysis helps identify relationships between numerical variables.

6. Conclusion

Exploratory Data Analysis helped summarize the Titanic dataset and identify patterns in passenger characteristics and survival outcomes.

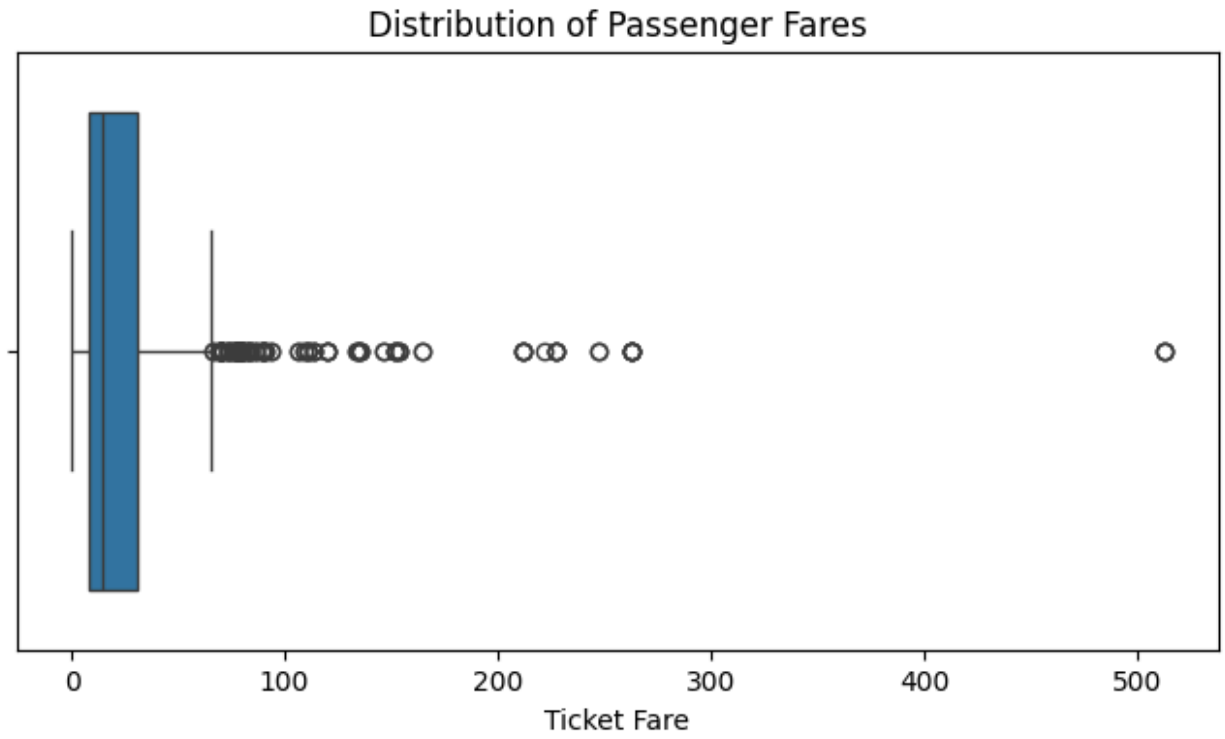
Visualizations:-

HISTOGRAM;-Age Distribution



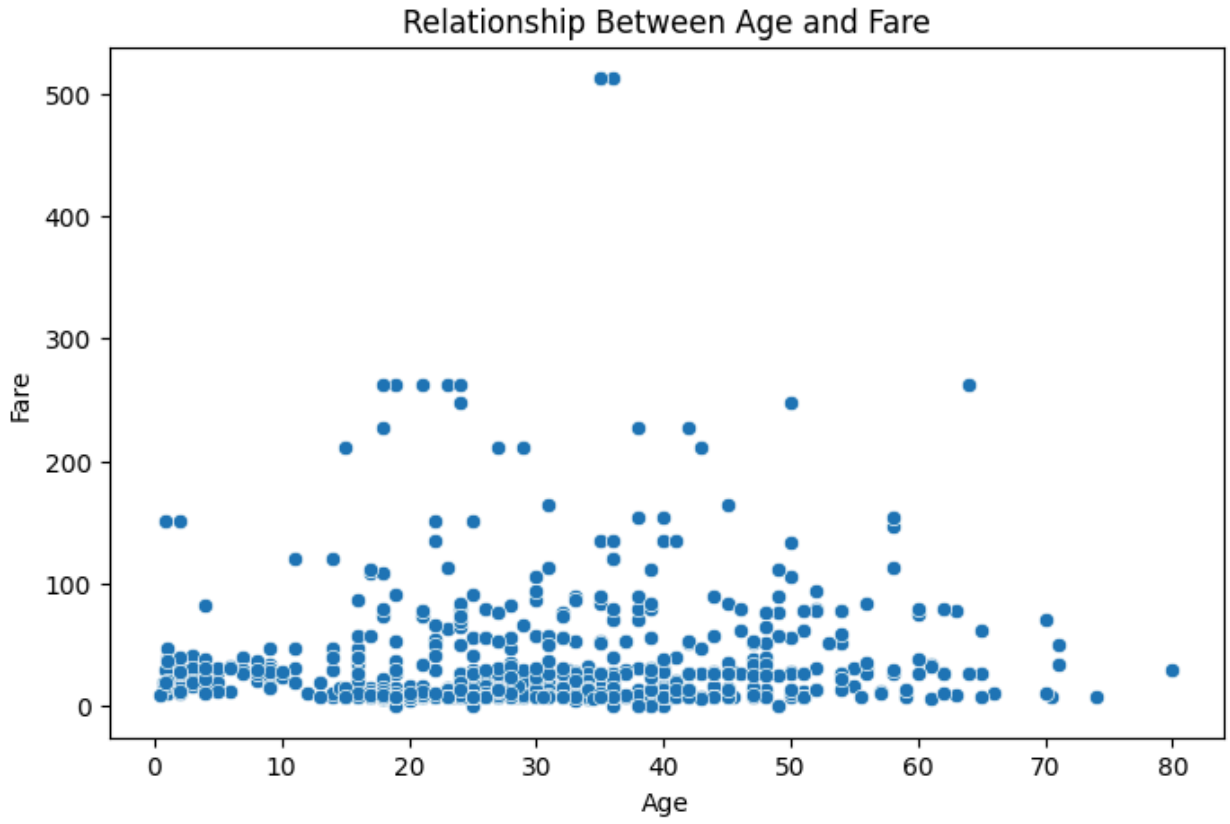
The histogram shows how passenger ages are distributed. Peaks indicate age ranges with more passengers.
The graph excludes rows with missing age values for this variable.

BOXPLOT:-Passenger Fares



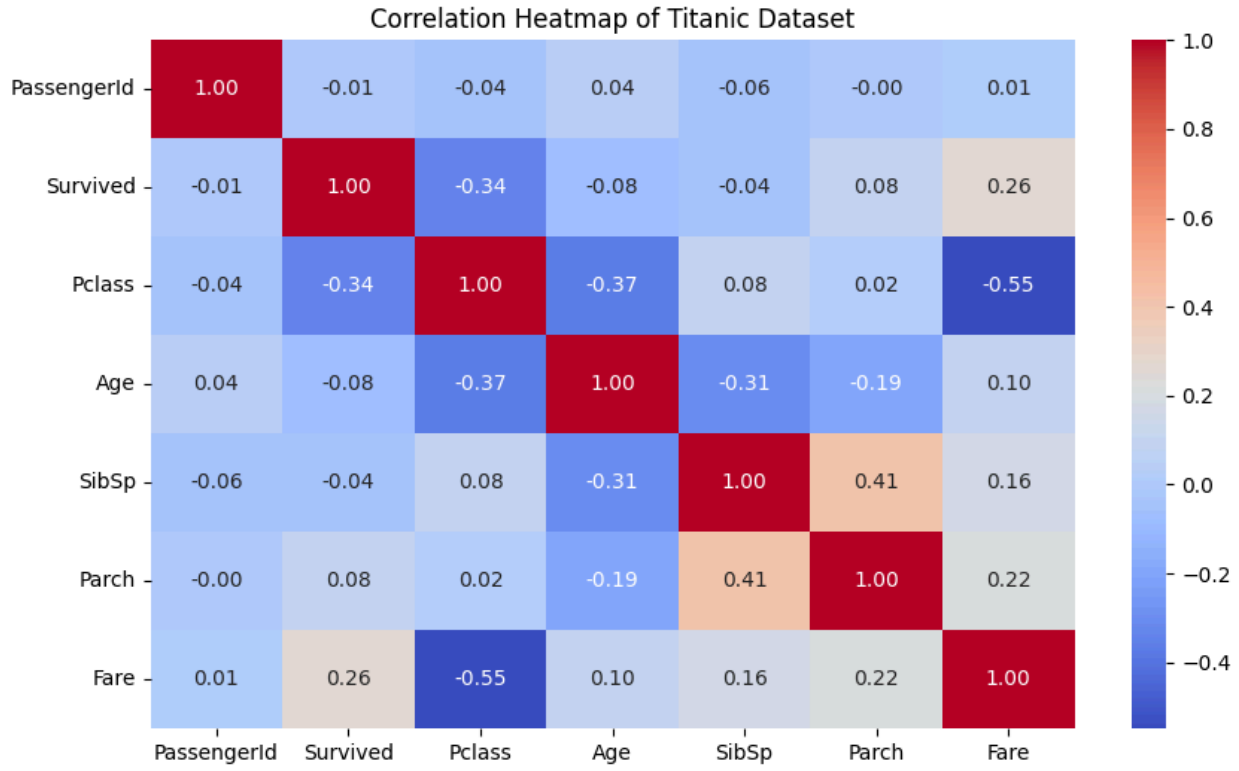
The boxplot summarizes the median, quartiles, spread, and potential outliers in ticket fares. Points or values
beyond the whiskers may indicate unusually high fares; they should be interpreted in context.

SCATTERPLOT:- Age versus Fare



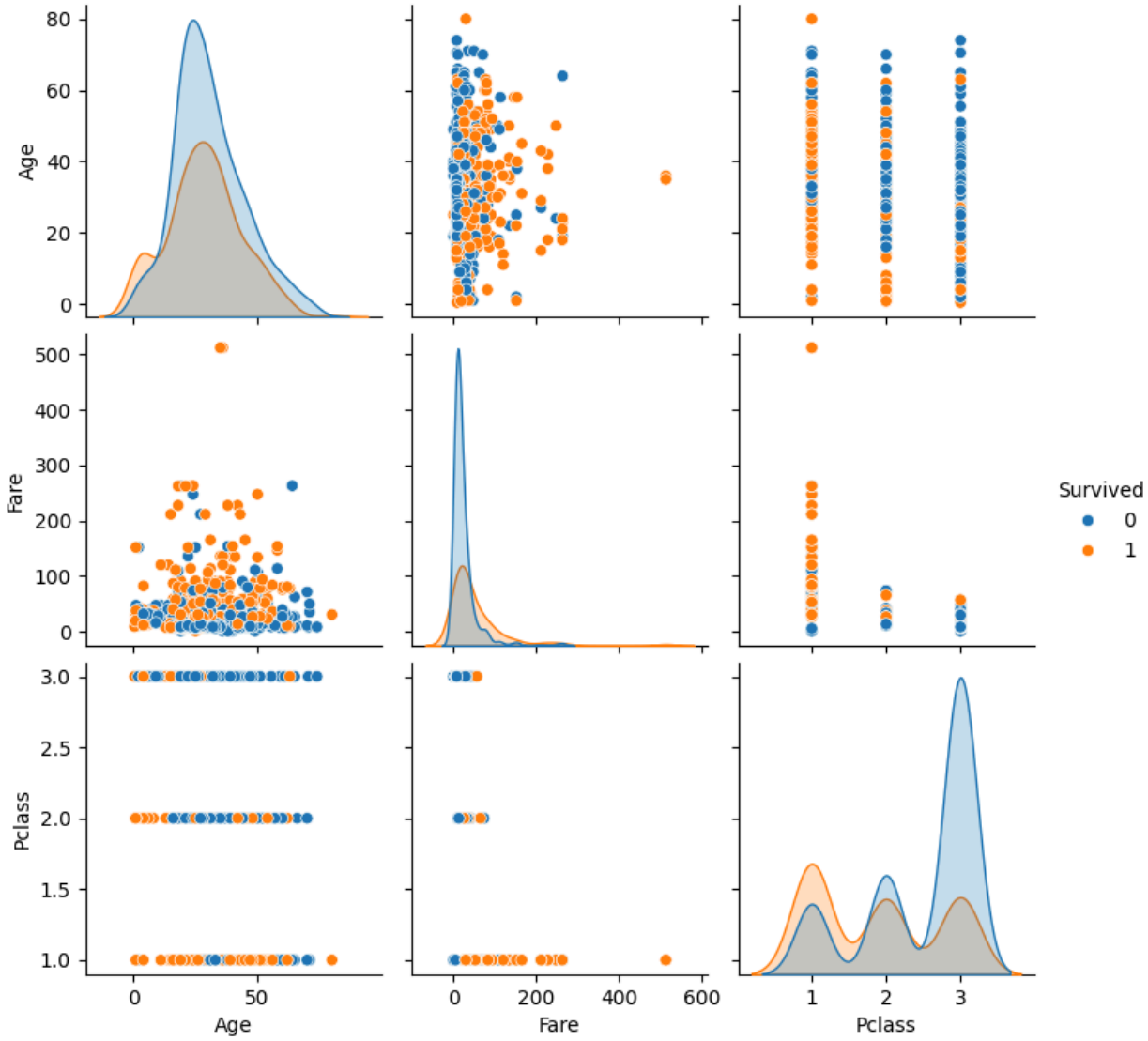
Each point represents a passenger with an available age and fare. The plot can reveal clusters and unusual values. A visible pattern should be described from the actual graph rather than assuming that age determines fare.

HEATMAP:- Correlations



The heatmap displays correlations between numeric columns. Values near +1 indicate positive linear
association, values near −1 indicate negative linear association, and values near 0 indicate little linear
association. Correlation does not prove causation.

PAIRPLOT:-Relationships between numerical variables



The pairplot compares pairwise distributions and relationships among the selected variables, with color
separating survival groups. It helps explore patterns but does not by itself establish why a passenger survived.